

beltvision: A Reproducible Classical Computer-Vision Inspection Engine for Conveyor Belts, Verified on Ground-Truth Synthetic Scenes

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Abstract—Conveyor-belt inspection asks a vision system to find the belt, measure its geometry, tell an empty return strand from a loaded one, and flag damage or foreign objects, and the honest difficulty is that most published demonstrations run on private field footage that no one else can rerun. **beltvision** is a reusable inspection engine that bundles the classical vision methods (CLAHE preprocessing, belt-edge geometry, semantic layering, granulometry, anomaly and tracking) behind a staged pipeline, and it ships something a field demo cannot: a deterministic suite of synthetic belt scenes carrying their exact ground truth, so the engine can be benchmarked with no external or private data. This note reports that benchmark. On six generated scenes spanning vertical, horizontal, diagonal, curved and laterally misaligned belts, the training-free classical chain recovers the belt-axis orientation to a mean error of 2.5° (worst 7.5° , all within the 8° product tolerance), recovers the belt footprint at a mean intersection-over-union of 0.81, segments the exposed belt rubber at 0.84 and the mineral load at 0.96, and recovers an injected lateral misalignment (-9° ground truth) to -7.5° , correct in sign and within 1.5° . It also reports the honest limitation the same ground truth exposes: the classical core does not isolate small foreign objects (per-class intersection-over-union 0.00), which is exactly the task the engine delegates to an optional, lazily imported learned lane. Every number regenerates from the seed and the committed code; no belt imagery is redistributed.

Index Terms—computer vision, conveyor belt, inspection, image segmentation, anomaly detection, granulometry, reproducibility, open-source software.

I. INTRODUCTION

A Conveyor belt carrying ore is a continuous inspection problem: a camera over the belt should locate the belt surface, measure its axis and alignment against the supporting structure, distinguish a loaded top strand from an empty return strand, estimate the fragment-size distribution of the load, and raise a flag on a rip, a hole or a tramp foreign object. Each of these has a mature classical-vision treatment, from CLAHE contrast equalisation [1] through morphological granulometry [2] to the deep anomaly-detection models that now lead the industrial benchmarks [5], [6]. The practical obstacle is not the methods; it is that a belt-inspection demonstration is almost always shown on a private recording from one plant, which means the numbers cannot be reproduced or compared by anyone else.

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Software note, 2026, for the **beltvision** package (v0.11.3, Apache-2.0): <https://github.com/fsantibanezleal/beltvision>.

beltvision is built to remove that obstacle. It is a library, not an application and not a dataset: a domain-agnostic inspection engine importable with a slim classical stack (`numpy`, `opencv`, `scikit-image` [3], `scipy`), whose heavier deep-learning engines are optional extras imported lazily and never at load time. Alongside the engine it ships a deterministic generator of synthetic belt scenes that carry their *exact* ground truth, the belt mask, the axis orientation, the medial-axis centreline, the injected misalignment, a four-class label map and the injected damage and foreign-object boxes, so the whole pipeline can run end to end and be scored against known geometry with no external data. This note is the benchmark that ground truth makes possible: how well the training-free classical chain recovers the belt geometry and semantics, across orientations, and where it honestly fails.

II. THE ENGINE

The engine is a frozen six-stage pipeline (`preprocess`, `feature_extraction`, `train`, `infer`, `evaluate`, `export`) whose stage signatures are stable while their bodies are the rework surface for heavier methods. Two data contracts bound it: an ingestion gate that accepts, flags or rejects an input frame, and a validated artifact manifest that a viewer can replay without rerunning the pipeline. A measured lane gate decides where a capability runs, in a browser, on a CPU server, or offline as a precompute artifact, from measured bytes and milliseconds rather than a hand-assigned label. The classical methods (preprocessing, belt-edge geometry, semantic layering, granulometry, tracking) run in the live lanes with the slim stack; the learned anomaly detectors (a convolutional auto-encoder, PaDiM [5] and PatchCore [6], evaluated under the MVTec AD protocol [4]) live behind the optional `[dl]` extra and are imported only inside the precompute lane. This note benchmarks the classical core, which is what runs without any model download or belt corpus.

III. GROUND-TRUTH GEOMETRY RECOVERY

The synthetic suite is six scenes chosen to break an all-vertical test: a vertical empty belt, a near-horizontal loaded belt, diagonals at 30° and 45° , a curved loaded belt, and a laterally misaligned belt. For each, the classical chain segments the belt, fits its principal axis, traces the centreline and measures the axis against the supporting structure, entirely

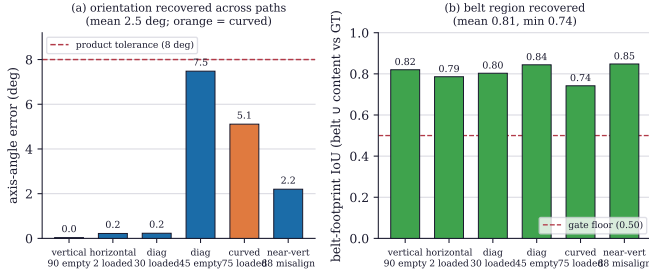


Fig. 1. The training-free classical chain (CLAHE, semantic layers, PCA axis, centreline, alignment) recovers the known geometry of the synthetic suite. (a) Belt-axis orientation error per scene, against the 8° product tolerance; the two larger errors are the empty diagonal and the curved belt (orange), where a single straight axis is an approximation by construction. (b) Belt-footprint intersection-over-union against the generated belt mask, well above the 0.50 recovery floor at every orientation.

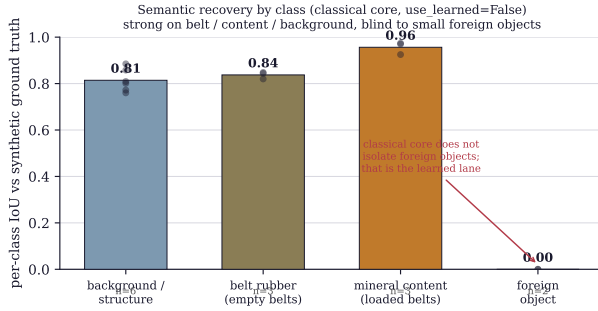


Fig. 2. Per-class segmentation of the classical core against the four-class synthetic ground truth, each class scored where it is the observable surface. Background/structure, exposed belt rubber and mineral content are recovered at intersection-over-union 0.81 to 0.96. Small foreign objects are not isolated by the classical core (per-class intersection-over-union 0.00): that class is the task the engine hands to its optional learned lane. Dots are per-scene values.

training-free (`use_learned=False`). Because the scene is generated, the recovered geometry is compared to the exact truth.

Fig. 1 reports the result. The belt-axis orientation is recovered to a mean error of 2.5° across the whole orientation span, with three of the six scenes below 0.25° ; the two larger errors, 7.5° on the empty 45° diagonal and 5.1° on the curved belt, are the expected cost of fitting a single straight axis to a low-contrast or genuinely curved path, and both remain inside the 8° tolerance the product documents. The belt footprint, the belt-and-content region against the generated belt mask, is recovered at a mean intersection-over-union of 0.81 (minimum 0.74 on the curved belt), comfortably above the 0.50 floor at every orientation. On the laterally misaligned scene, whose ground-truth belt-versus-support misalignment is -9° , the chain measures -7.5° : correct in sign and within 1.5° of the truth, the reading a belt-tracking alarm would key on. The whole classical chain runs in a mean of 71 ms per frame on one CPU core.

IV. SEMANTIC RECOVERY, AND THE HONEST LIMITATION

The same ground truth carries a four-class label map (external structure, belt rubber, mineral content, foreign object), so the semantic layering can be scored per class (Fig. 2). Scored

where each class is the exposed surface, the classical core recovers the background and support structure at intersection-over-union 0.81, the exposed belt rubber on empty belts at 0.84, and the mineral load on loaded belts at 0.96; on a loaded belt the rubber is correctly relabelled as content rather than lost, which is why the belt class is scored on the empty belts where the rubber is actually visible.

The honest result is the fourth bar. The classical core does not isolate the injected foreign objects: their per-class intersection-over-union is 0.00, because a small bright tramp object on a textured belt is not separable by contrast and morphology alone, and the pixels fall into the surrounding content or belt class. This is not a bug to paper over; it is the reason the engine keeps a learned anomaly lane at all. The classical core is the fast, dependency-light path that recovers belt geometry and coarse content, and the learned lane (PaDiM [5], PatchCore [6], trained normal-only under the MVTec AD discipline [4]) is the path for the foreign-object and fine-anomaly task the classical core cannot do. Reporting the 0.00 rather than hiding it is what makes the split defensible.

V. SCOPE AND REPRODUCTION

The scope is stated plainly. `beltvision` is a library other applications consume, not a service and not a trained-model zoo; the bundled cases are a registry plus deterministic synthetic scenes, and real frames are brought by the consumer. The benchmark here is therefore a *geometry-recovery* benchmark on synthetic scenes with known truth, not a field-performance claim on plant footage: it proves that the classical chain recovers orientation, footprint, content and misalignment across orientations, and it locates the one class the classical core cannot do. The learned anomaly lane is described but not benchmarked here, because a fair learned benchmark needs a real belt corpus, which the library deliberately does not bundle or redistribute. Everything in Fig. 1 and Fig. 2 regenerates from the seed and the committed code: `run_bench.py` runs the synthetic suite through the classical chain and writes the measured errors, and `make_figs.py` renders the figures; no belt imagery is involved.

VI. CONCLUSION

`beltvision` packages the classical conveyor-belt vision methods behind a staged, contract-bounded pipeline and, unusually, ships the ground truth to test itself. On a six-scene synthetic suite spanning orientations, its training-free classical core recovers belt orientation to 2.5° mean error, the belt footprint to intersection-over-union 0.81, mineral content to 0.96, and a lateral misalignment to within 1.5° of a -9° truth, at 71 ms per frame on a CPU. It also reports, from the same ground truth, that the classical core cannot isolate small foreign objects, which is the task delegated to an optional learned lane. A reproducible, data-free benchmark of a real inspection engine, honest about what the classical path does and does not recover, is the note's contribution.

APPENDIX A REPRODUCTION

The committed benchmark (manuscripts/belt-inspection/data/bv.json) carries, per scene, the ground-truth and recovered belt orientation, the belt-footprint and per-class intersection-over-union, the measured misalignment, the centreline root-mean-square error on straight belts, and the per-frame CPU time. `figures/run_bench.py` produces it by running `beltvision.cases.synthetic`'s ground-truth suite through `compute_layers` and `compute_belt_geometry` with `use_learned=False`; `figures/make_figs.py` renders both figures from it. The classical stack is `numpy/opencl/scikit-image/scipy`; no model artifacts or belt corpus are needed to reproduce any number in this note.

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